Course Name and Code	*Data Security - ITI410*
Number	*1*
Title	*OpenSSL*
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Classes	*C8, C9 and C10*
Start Date	*November 20th, 2012*
Delivery Date	*December 31st, 2012*
Resources	*Cryptography Techniques and Applications*

OBJECTIVES

The assignment aims at enforcing the cryptography techniques. It helps the student in acquiring the necessary knowledge in cryptography applications and in exchanging secure data using encryption techniques used by SSL (Secure Socket Layer) protocol.

Assignment objectives can be summarized as:

- Understanding cryptography applications.
- Using digital certificates.
- Configuring and launching SSL servers.
- Configuring and using SSL clients.
- Using SSL to secure Web browsing.

The assignment aims at making the student familiar with certain security techniques and applications; essentially asymmetric cryptography, certificates and SSL. The student will be able to employ these services in a scenario of secure Web browsing.

This assignment is based on the OpenSSL toolkit. OpenSSL is a widely-used open-source, cryptography API.

ASSIGNMENT DETAILS

1. Purpose

The purpose is to master OpenSSL, an open source toolkit providing general purpose cryptography features. It aims at mastering the following issues:

1. Creating key exchange parameters.
2. Creating public/private keys.
3. Creating, auto-signing and signing certificates.
4. Configuring and launching SSL servers and clients.
5. Securing Web data over SSL.

To achieve these goals, it is required to build a Certification Authority (CA). The CA responsibilities are:

- Creating and signing its own certificate.
- Signing certificates for customers.

It is also required to use certificates and key-exchange parameter to configure and run SSL servers and clients. Finally, it is required to browse Web data securely using SSL clients.

2. Functionalities

2-1. Creating Certificates

You will create a Certification Authority (CA), its auto-signed certificate and the required certificates for SSL server and client.

You need first to create a subfolder called "`newcerts`", a text file called "`serial.txt`" containing the text "01" and an empty text file called “`index.txt`".

1. Write a command line in a batch file called "`Generate_CAKeys.bat`" to generate an RSA public/private key pair for the Certification Authority (CA). Run the command and save the result in a file called "`ca.key`".
2. Write a command line in a batch file called "`Create_CACert.bat`" to create the auto-signed certificate of the CA using the configuration file "`openssl.txt`" provided with the assignment. Use the name "`svuonline.org`" for the CA in its certificate. Run the command and save the result in a file called "`ca_cert.pem`".
3. Write a command line in a batch file called "`Visualize_CACert.bat`" to create a readable version out of "`ca_cert.pem`" to visualise the content of the auto-signed certificate of the CA. Run the command and save the result in a file called "`ca_cert_content.txt`".
4. Write two command lines in a batch file called "`Generate_UserKeys.bat`" to generate RSA public/private key pairs for the SSL server and client. Run the commands and save the results in two files; one is called "`server.key`" for the SSL server, and the other is called "`client.key`" for the SSL client.
5. Write two command lines in a batch file called "`Generate_UserRequests.bat`" to generate a certificate requests for the SSL server and client using the configuration file "`openssl.txt`" provided with the assignment. Run the commands and save the results in two files; one is called "`server.crs`" for the SSL server, and the other is called "`client.crs`" for the SSL client. Use the name "`SSL Server`" for the server in its certificate request, and the name "`SSL Client`" for the client in its certificate request.
6. The certificate requests should be sent to the CA to be signed, which should generate the required certificates. Write two command lines in a batch file called "`Sign_UserRequests.bat`" to sign both requests "`server.crs`" and "`client.crs`", in order to generate their corresponding certificates. Run the commands and save the results in two files; one is called "`server_cert.pem`" for the SSL server, and the other is called "`client_cert.pem`" for the SSL client.

2-2. SSL Client / Server

You will create and launch three types of SSL servers on three different ports. Two of the SSL servers emulate two different simple Web servers. You will create and launch three SSL clients to test the three different SSL servers. You will also use a traditional Web browser over SSL and compare it with the SSL client you create and use to send HTTP commands.

7. Write a command line in a batch file called "`Create_DHParam.bat`" to create the Diffie-Hellman parameters that will be used by the SSL server to share secret keys with its clients. Run the command and save the result in a file called "`dhparam.pem`", in which the parameters should be readable.
8. Write a command line in a batch file called "`Create_SSLServer01.bat`" to create an SSL server that listens on the port 1234, and that uses the previously-created Diffie-Hellman parameters saved in "`dhparam.pem`". You should use the previously-created server certificate saved in "`server_cert.pem`". This server does not request certificates from its clients, and does not emulate a Web server. Run the command to launch this first SSL server.
9. Write a command line in a batch file called "`Create_SSLClient01.bat`" to create an SSL client that connects to the first SSL server on the port 1234, and verifies the server certificate chain with a depth of 2. Save the output messages of the command to the file "`client01.txt`". Run the command to launch this first SSL client.
10. Write a command line in a batch file called "`Create_SSLServer02.bat`" to create an SSL server that listens on the port 1235, and that uses the previously-created Diffie-Hellman parameters saved in "`dhparam.pem`". You should use the previously-created server certificate saved in "`server_cert.pem`". This server does not request certificates from its clients, but it emulates a simple Web server. Run the command to launch this second SSL server.
11. Create a simple Web page called "`mygroup.html`" providing for each one in your assignment group the name as a heading, the SVU ID and the email address as a link in a paragraph, and in a separate paragraph the URL you should use to open this Web page in a Web browser on the same machine of the previous SSL-based simple Web server. Use only HTML to create this Web page. Save the file of the Web page to the default root of the previous SSL-based simple Web server.
12. Write a command line in a batch file called "`Create_SSLClient02.bat`" to create an SSL client that connects to the second SSL server on the port 1235, and verifies the server certificate chain with a depth of 2. Save the output messages of the command to the file "`client02.txt`". Run the command to launch this second SSL client, and use on it the "`GET`" command of HTTP v1.0 to get the Web page "`mygroup.html`". Add the HTTP "`GET`" command you used to the file "`mygroup.html`" in a separate paragraph.
13. Write a command line in a batch file called "`Create_SSLServer03.bat`" to create an SSL server that listens on the port 1236, and that uses the previously-created Diffie-Hellman parameters saved in "`dhparam.pem`". You should use the previously-created server certificate saved in "`server_cert.pem`". The clients of this server must supply certificates with

a verification depth of 2, and it emulates a simple Web server. Run the command to launch this third SSL server.

14. Use a Web browser to ask the previous SSL-based simple Web server for the Web page "`mygroup.html`" . Add the URL you used to the file "`mygroup.html`" in a separate paragraph. Explain the reaction of the SSL-based Web server in this case in a document called "`SSLServer03.doc`".
15. Write a command line in a batch file called "`Create_SSLClient03.bat`" to create an SSL client that connects to the third SSL server on the port 1236, verifies the server certificate chain with a depth of 2, and sends its own certificate to the SSL server. You should use the previously-created client certificate saved in "`client_cert.pem`". Save the output messages of the command to the file "`client03.txt`". Run the command to launch this third SSL client, and use on it the "GET" command of HTTP v1.0 to get the Web page "`mygroup.html`". Add an explanation of the reaction of the SSL-based Web server in this case to the document called "`SSLServer03.doc`".

DELIVERABLES

The student should deliver a one-folder package containing – in addition to the subfolder "`newcerts`", the text file "`serial.txt`", the text file "`index.txt`" and the configuration file "`openssl.txt`" – the following files:

1. `Generate_CAKeys.bat`
2. `ca.key`
3. `Create_CACert.bat`
4. `ca_cert.pem`
5. `Visualize_CACert.bat`
6. `ca_cert_content.txt`
7. `Generate_UserKeys.bat`
8. `server.key`
9. `client.key`
10. `Generate_UserRequests.bat`
11. `server.crs`
12. `client.crs`
13. `Sign_UserRequests.bat`
14. `server_cert.pem`
15. `client_cert.pem`
16. `Create_DHParam.bat`
17. `dhparam.pem`
18. `Create_SSLServer01.bat`
19. `Create_SSLClient01.bat`
20. `client01.txt`
21. `Create_SSLServer02.bat`
22. `mygroup.html`
23. `Create_SSLClient02.bat`
24. `client02.txt`
25. `Create_SSLServer03.bat`
26. `SSLServer03.doc`
27. `Create_SSLClient03.bat`
28. `client03.txt`

ASSIGNMENT DELIVERY

1. Students can work in groups of four at maximum. Student names in each group should be communicated to the tutor prior to assignment delivery.

2. A ZIP package should be delivered. The file will be named as follows:
   `Assignment_ITI410_S12_<Student1ID>_<Student2ID>.zip`
   For example:
   `Assignment_ITI410_S12_mahmoud_13452_jameel_16272.zip`
   `Assignment_ITI410_S12_mouna_17499.zip`
3. Assignment delivery through Email is NOT accepted.

ASSIGNMENT SOURCE AND RESOURCES

1. The assignment source will be sent to students by email and it can be downloaded from Moodle.
2. The OpenSSL package to be used is "Win32OpenSSL-1_0_1c.exe" and it can be downloaded from:
   http://slproweb.com/download/Win32OpenSSL-1_0_1c.exe
3. To read about OpenSSL go to the official website http://www.openssl.org/